

Bidding for Incomplete Contracts: An Empirical Analysis of Adaptation Costs[†]

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Procurement contracts are often renegotiated because of changes that are required after their execution. Using highway paving contracts we show that renegotiation imposes significant adaptation costs. Reduced form regressions suggest that bidders respond strategically to contractual incompleteness and that adaptation costs are an important determinant of their bids. A structural empirical model compares adaptation costs to bidder markups and shows that adaptation costs account for 7.5–14 percent of the winning bid. Markups from private information and market power, the focus of much of the auctions literature, are much smaller by comparison. Implications for government procurement are discussed. (JEL D44, D82, D86, H57, L13, L74, R42)

The benefits of using auctions to procure goods and services are well known and vigorously advocated. Competitive bidding results in low prices and limits favoritism. Employing auctions to procure standard goods such as pencils, printers, and bookkeeping software is straightforward, yet procuring custom made goods and services to fit a buyer's unique needs poses challenges. First, the buyer must spend resources to translate operational needs into well defined and communicable specifications. Second, needs for adaptations and changes often result from inadequate designs and specifications, changes in the external environment, or more generally, the extent to which the initial contract is incomplete.

Incomplete contracts force the buyer and supplier to negotiate adaptations both to the scope of work and to compensation, which may result in considerable discrepancies between the winning bid and the final payment. A well known example is the “Big Dig” highway artery in Boston, for which 12,000 changes to more than 150 contracts led to \$1.6 billion in cost overruns, most of which can be traced back

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to unsatisfactory design.¹ One source of cost overruns is the additional work that was not anticipated. In addition, *adaptation costs* are incurred by disruptions to the normal flow of work that could have been avoided with adequate planning in advance. Renegotiating the contract also generates adaptation costs in the form of haggling, dispute resolution, and opportunistic behavior.

Despite the prevalence of incomplete contracts, their effect on procurement in general, and on adaptation costs in particular are ignored almost without exception in both the theoretical and empirical auction literatures. This paper contributes by offering a first attempt to measure the economic costs of ex post adaptations that result from incomplete contracts. We develop a simple framework of bidding for incomplete contracts and apply it to highway procurement in the state of California. Our analysis suggests that adaptation costs are large, imposing significant extra costs on public procurement.

Our approach is guided by three important features of highway procurement. First, given the competitive nature of the highway construction industry (publicly traded firms in our sample report profit margins of less than 3 percent), bidders must anticipate ex post changes and try to include any adaptation costs in their bids. Second, highway procurement uses “unit-price auctions” where contracts are summarized by a list of estimated input quantities required to do the job. A bidder submits a list of itemized (unit) prices, which multiplied by estimated quantities result in the bidder’s total bid. The contractor with the lowest total bid wins the auction. Third, the winning total bid is seldom equal to the final compensation because ex ante *estimated* quantities and ex post *actual* quantities never perfectly agree. Furthermore, *changes in scope* may be necessary when some specifications of the project need to be altered.

We develop a stylized model where contractors form rational expectations about the ways in which actual quantities will differ from estimated ones, as well as whether changes in scope will be required. Hence, expected changes in payments and adaptation costs will be incorporated into the bids ex ante.² We then apply the empirical framework of our model to a panel dataset of highway contract bids that we have collected from Caltrans (California’s Department of Transportation). The data includes bidder identities, bids, detailed cost estimates, and measures of cost advantages. Unlike most studies of procurement, our data also contains detailed information on how the initial designs were altered, including both *estimated* and *actual* quantities for all work items in the contract, as well as payments to the contractor from *changes in scope*.

Our empirical analysis first presents reduced form estimations where the strategy for identifying adaptation costs is based on our theoretical model. Suppose that the contractors all expect additional payments due to ex post changes. Controlling for production costs, if there were no adaptation costs then competition implies that for every extra dollar of expected profits, each contractor should lower its bid by exactly

¹ According to the *Boston Globe*, “[a]bout \$1.1 billion of that can be traced back to deficiencies in the designs, records show: \$357 million because contractors found different conditions than appeared on the designs, and \$737 million for labor and materials costs associated with incomplete designs.” See http://www.boston.com/news/specials/bechtel/part_1/.

² This is in line with Haile (2001) who explores timber auctions where forward looking rational bidders take into account the future possibility of resale to calculate their optimal bid.

one dollar. If the regression is correctly specified, the coefficient on ex post additional payments should therefore be -1 . We find that some coefficients are closer to $+1$, implying that ex post changes on net generate more costs than revenue.

A concern with our identification strategy is that complex projects, for which ex post changes are likely, also have higher production costs. This would confound adaptation costs with production costs and curtail our results. We address this concern using an instrumental variables strategy and identify an exogenous shifter of ex post payments that is uncorrelated with project characteristics. As we explain in Section IVC, we use the identity of the Caltrans engineer assigned to the project as an instrument for changes to compensation. Individual engineers have discretion over ex post adjustments, making some engineers easier to work with while others are known to cause problems. Engineers are randomly assigned to manage several projects and their identities are known to contractors at the time of bidding, allowing us to identify adaptation costs using engineer assignments.

In order to quantify the magnitude of adaptation costs as a markup over production costs, we estimate a structural model that builds on our theoretical analysis and decomposes markups into three components. First, as in any auction model, markups are a function of private information and local market power. Second, as in Athey and Levin (2001), who investigate behavior in unit-price timber auctions, we estimate the impact of “unbalanced” bids. Namely, contractors can increase expected profits by increasing (decreasing) unit prices on items that are expected to overrun (underrun). Third, we estimate the adaptation costs from changes to the initial specifications (estimated quantities) to uncover the ex post costs of misspecified ex ante designs. The structural model estimates are consistent with the reduced form findings and show that adaptation costs are larger than the other two markup drivers. We conclude our empirical analysis with a conservative bounding strategy to find upper and lower bounds on the adaptation costs. We continue to find large and significant estimates of adaptation costs under these two specifications, and conclude that our estimates suggest that adaptation and changes are a major determinant of bids in this industry and an important potential source of inefficiency.

While largely ignored in the empirical procurement literature, concerns over the impact of adaptation costs are prevalent in the construction management literature (see Hinze 1993; Clough and Sears 1994; and Sweet 1994). The earliest foundations of transaction costs economics (Williamson 1971) argue that incomplete contracts imply both needs for adaptation and potential frictions. This idea has been explored in the procurement literature, and several studies emphasize the importance of adaptation costs including Crocker and Reynolds (1993); Bajari and Tadelis (2001); and Chakravarty and MacLeod (2009). Our paper contributes to this literature because, to the best of our knowledge, there are no empirical estimates of the *dollar value* of these costs. We demonstrate that standard methods for estimating auctions can be modified to yield an estimate of adaptation costs in procurement auctions.

This paper also contributes to the literature on structural estimation of auctions (see, e.g., Paarsch 1992; Guerre, Perrigne, and Vuong 2000; and Krasnokutskaya 2011). First, our estimates imply that adaptation costs, which have until now been largely ignored, seem to impose more distortions and frictions as compared with market power and unbalanced bidding. Second, our results suggest that profit markups in standard bidding models are often misspecified because they do not account

Item	Description	Estimated quantity	Per unit bid	Estimated item bid
1.	Asphalt (tons)	25,000	\$25.00	\$625,000.00
2.	Sidewalk (square feet)	10,000	\$9.00	\$90,000.00
3.	Rumble strips	50	\$5.00	\$250.00
Total bid:				\$715,250.00

FIGURE 1. UNIT PRICE CONTRACT: AN EXAMPLE

for the discrepancies between initial bids and final payments, omitting important parts of a contractor's revenues and costs. This in turn implies that policies geared toward reducing the amount of contractual incompleteness may have large benefits by reducing the costs of public procurement.

I. Competitive Bidding for Highway Contracts

Highway construction, as well as other public sector procurement, is often done using unit-price contracts. (See, e.g., Hinze 1993.) Government engineers first prepare a list of items that describe the tasks and materials required for the job. In the contracts we study, items include laying asphalt, installing new sidewalks, and striping the highway. For each work item, engineers provide an estimate of the quantity needed to complete the job. For example, they might estimate 25,000 tons of asphalt, 10,000 square yards of sidewalk, and 50 rumble strips. The itemized list is publicly advertised along with a detailed set of plans and specifications that describe how the project is to be completed.

An interested contractor will bid per unit prices for each work item on the engineer's list. Figure 1 shows an example of the structure of a completed bid, which must be sealed and submitted prior to a set date. When the bids are opened, the contract is awarded to the contractor with the lowest estimated total bid, defined as the sum of the estimated individual line item bids.³

Actual quantities will almost always be different from estimated quantities. The difference may be substantial if there are unexpected conditions or work has to be redone or eliminated. As a result, final payments made to the contractor are almost never equal to the original bid, and their determination can be complicated. Caltrans' *Standard Specifications* and its *Construction Manual* discuss the determination of the final payment. To a first approximation, there are three primary reasons for modifying the payments away from the simple sum of actual unit costs.

First, if the difference between estimated and actual quantities is large, or if it is thought to be due to negligence by one party, both sides will renegotiate an *adjustment of compensation*.⁴ Using our example, if the asphalt ran over by 10,000 tons, Caltrans would hesitate to pay an extra \$250,000, and the parties may negotiate an adjustment to lower the total bill. In our data, adjustments are recorded as a lump

³The lowest bid can be rejected if the bidder is not appropriately bonded or does not have a sufficient amount of work awarded to disadvantaged business enterprises as subcontractors. Also, bids judged to be highly unbalanced can be rejected, as discussed further below.

⁴In the particular case of highway construction procured by Caltrans, this type of adjustment is called for if the actual quantity of an item varies from the engineer's estimate by 25 percent or more.

sum change, but this may also be a way for parties to adjust the implied unit price on a particular item.

Second, there may be a *change in scope*. For instance, the original scope might be to resurface two miles of highway. However, the engineers and contractor might discover that the subsurface is not stable and that certain sections need to be excavated and have gravel added, an activity that was not originally described. In most cases, the contractor and Caltrans will negotiate a *change order* that amends the scope of the contract as well as the final payment. If negotiations break down, this may lead to arbitration or a lawsuit. Payments from changes will appear in two ways. One is that changes in the actual ex post quantities of prespecified items will be compensated for through the unit prices. Another is by extra payments made to reflect the use of unanticipated materials or other adjustment costs, and they are recorded as *extra work* in our data.

Finally, the payment may be altered because of *deductions*. If work is not completed on time or if it fails to meet specifications, Caltrans may deduct liquidated damages. Such deductions are often a source of disputes between Caltrans and the contractor. The contractor may argue that the source of the delay is poor planning or inadequate specifications provided by Caltrans, while Caltrans might argue that the contractor's negligence is the source of the problem. The final deductions imposed may be the outcome of heated negotiations or even lawsuits and arbitrations between contractors and Caltrans.

It is widely believed in the industry that some contractors attempt to strategically manipulate their bids in anticipation of changes to the payment. Contractors read the plans and specifications to forecast the likelihood and magnitude of changes to the contract. For instance, consider the example of Figure 1, in which the total bid is \$715,250. Suppose that after reading the plans and specifications, the contractor expects asphalt to overrun by 5,000 tons and sidewalk to underrun by 3,000 square feet. If he changes his bid on sidewalk to \$5.00 and his bid on asphalt to \$26.60 then his total bid will be unchanged. However, this will increase the contractors' expected total payment to \$833,750.00 ($26.6 \times 30,000 + 5 \times 7,000 + 5 \times 50$) compared to \$813,750.00 when bids of \$25.00 and \$9.00 are entered. This increases his total payment *without increasing his total bid*, fixing his probability of winning. A bid is referred to as *unbalanced* if it has unusually high unit prices on some items (expected to overrun) and unusually low unit prices on others (expected to underrun).

Athey and Levin (2001) note that the optimal strategy for a risk neutral contractor is to submit a bid that has zero unit prices for some items that are overestimated, and put all the actual costs on items that are underestimated. In the data, however, while zero unit price bids have been observed, they are very uncommon. Athey and Levin suggest that one reason for this is risk aversion. After speaking with industry participants and reading industry sources we believe that for construction contracts other considerations are more important. In particular, Caltrans is not required to accept the low bid if it is deemed to be irregular (see Sweet 1994 for an in depth discussion of irregular bids). A highly unbalanced bid is a sufficient condition for a bid to be deemed irregular. As a result, a bid with a zero unit price is very likely, if not certain, to be rejected.⁵

⁵Using Blue Book prices and previous bids, Caltrans is able to check whether bids for certain work items are unusually high or low. In our data, 3 percent of the contracts are not awarded to the low bidder, and according to industry sources the mostly likely reason is unbalanced bids.

Also, the *Standard Specifications* and the *Construction Manual* indicate that unit prices on items that overrun by more than 25 percent are open to renegotiation. In these negotiations, Caltrans engineers will attempt to estimate a fair market value for a particular unit price based on bids submitted in previous auctions and other data sources. Caltrans may also insist on renegotiating unit prices even when the overrun is less than 25 percent if the unit prices differ markedly from estimates. This suggests that there are additional limitations on the benefits of submitting a highly unbalanced bid.

II. Bidding for Incompletely Specified Contracts

In this section we use the factual descriptions above to develop a simple variant of a standard private values auction model that will be the basis for our empirical models.

A. Basic Setup

A project is characterized by tasks, $t = 1, \dots, T$ and a vector of estimated quantities, $\mathbf{q}^e = (q_1^e, \dots, q_T^e)$ for each task, and is communicated by the buyer to $m > 1$ commonly known bidders. The actual ex post quantities that will be needed to complete the task is given by $\mathbf{q}^a = (q_1^a, \dots, q_T^a)$, and is independent of the bidder who is selected to perform the work.

Since we focus on ex post adaptation costs and not ex ante private information rents, we assume an extreme form of asymmetric information between the buyer and the bidders. We assume that each bidder has perfect foresight about the actual quantities $\mathbf{q}^a$ while the buyer (Caltrans) is unaware of $\mathbf{q}^a$ and only considers $\mathbf{q}^e$. Perfect foresight can naively be interpreted as the contractors knowing the actual $\mathbf{q}^a$. By assuming that contractors are risk neutral, a more convincing interpretation is that contractors do not have *exact* information about $\mathbf{q}^a$, but instead have *symmetric uncertainty* about the actual quantities, resulting in common rational expectations over actual quantities. This interpretation is useful for the empirical model because it generates a source of noise that is not specific to the contractor's information or the observable project characteristics.

Despite having symmetric information about $\mathbf{q}^a$, bidders differ in their private information about their own costs of production. Let c_t^i denote bidder i 's per unit cost to complete task t and let $\mathbf{c}^i = (c_1^i, \dots, c_T^i) \in \mathbb{R}_+^T$. The total cost to i for installing the vector of quantities $\mathbf{q}^a$ will be $\mathbf{c}^i \cdot \mathbf{q}^a$, the vector product of the costs and the actual quantities. The costs (type) of contractor i are drawn from a well behaved joint density $f_i(\mathbf{c}^i)$ with support on a compact subset of $\mathbb{R}_+^T$. The distributions are common knowledge, but only contractor i knows $\mathbf{c}^i$. Also, costs are independently distributed conditional on publicly observed information.⁶ Hence, bidders have symmetric rational expectations about what needs to be done but they have asymmetric private information about the costs of production.

⁶Private values is the common assumption for this industry (e.g., Porter and Zona 1993; Krasnokutskaya 2011). Common values with multiple units is intractable and is beyond the scope of this paper.

Bidder i submits a unit price vector $\mathbf{b}^i = (b_1^i, \dots, b_T^i)$ where b_t^i is his unit bid on item t . The *score* of bidder i as his total bid $s^i = \mathbf{b}^i \cdot \mathbf{q}^e$, and i wins the auction if and only if $s^i < s^j$ for all $j \neq i$. Hence, bidders participate in a simple linear scoring rule auction where each bid vector is transformed into a score, the estimated price.

Bidder i 's total cost of producing actual quantities $\mathbf{q}^a$, which we refer to as his *type*, is denoted by $\theta^i \equiv \mathbf{c}^i \cdot \mathbf{q}^a$. Let $R(\mathbf{b}^i)$ be the gross revenue that bidder i expects to receive if he wins with a bid of $\mathbf{b}^i$. His expected profit is given by

$$\pi_i(\mathbf{b}^i, \theta^i) = (R(\mathbf{b}^i) - \theta^i)(\Pr\{s^i < s^j \text{ for all } j \neq i\}).$$

B. Revenues and Adaptation Costs

If the only source of revenue was determined by unit prices and actual quantities then revenues would equal $\mathbf{b}^i \cdot \mathbf{q}^a$. As discussed earlier, however, gross revenue is affected by adjustments (A), extra work (X), and deductions (D). As with actual quantities, we assume that these three components are independent of which bidder wins the contract, and that bidders have no control over them.

Because contractors are risk neutral and have symmetric rational expectations about adjustment costs, we introduce each of these three components as expected values, and include them additively into the bidders' profit function. In the absence of adaptation costs, the revenues to the winning bidder i are equal to

$$R(\mathbf{b}^i) = \sum_{t=1}^T b_t^i q_t^a + A + X + D.$$

Any payments captured by $A + X + D$ are just a transfer of funds from the buyer to the contractor. However, in the presence of adaptation costs every dollar that is transferred has less than its full impact on profits.

As discussed earlier, there are two kinds of adaptation costs. The first are *direct adaptation costs* due to disruption of the originally planned work. Changes can disrupt the efficient rhythm of work, and it is not unusual for changes to cut in half the amount of asphalt laid by a contractor in a day. The project may take twice as long to complete and perhaps double the labor and capital costs.⁷ A second source of adaptation costs are *indirect adaptation costs* due to resources devoted to contract renegotiation and dispute resolution.⁸ Each side may try to blame the other for any needed changes, and they may disagree over the best way to change the plans and

⁷ An example witnessed by one of the authors occurred while overlaying a concrete highway with asphalt where innumerable cracks had been patched with a dark, black "latex joint sealer." As paving began, the latex came in contact with hot asphalt, and the heated joint sealer would often explode through the freshly laid mat of asphalt. As a result, the latex joint sealer had to be removed from thousands of cracks by laborers using mostly hand tools before state engineers would allow the contractor to overlay the existing concrete road. This greatly slowed down the rate at which paving could occur, causing trucks to frequently stand in line for an hour before they could dump their asphalt into the paver. The contractor and the state engineers disagreed vehemently about the additional expense caused by the need to remove the crack sealer. Compensation for this change had to be renegotiated at length.

⁸ Estimates place the value of change orders at \$13 to \$26 billion per year, but researchers have noted that with the additional costs related to filing claims and legal disputes, the total cost of changes could reach \$50 billion annually (see Hanna and Gunduz 2004).

specifications. Disputes over changes may generate a breakdown in cooperation on the project site and possibly lawsuits.⁹

In reality, the contractual incompleteness that leads to adjustments, extra work, and deductions will be positively correlated with the direct costs from disrupting the normal flow of work and the indirect costs of renegotiation. We assume that these extra costs are proportional to the size of adjustments, extra work, and deductions. For example, the imposed adaptation cost from extra work X is given by $\tau_x X$.

It is useful to distinguish between positive and negative ex post adjustments to revenues. By definition, any extra work adds compensation to the contractor while any deduction reduces the contractor's compensation. This implies that $X > 0$ and $D < 0$. The adjustments A , however, can be positive or negative. We separate these so that positive (negative) adjustments are labeled $A_+ > 0$ ($A_- < 0$). For positive ex post income, adaptation costs cause surplus to be dissipated while for negative ex post income, adaptation costs cause the contractor to suffer a loss above and beyond the actual loss imposed by the adjustments or deductions. The (positive) coefficients τ will measure these extra losses. Thus, we can write down the total ex post costs of adaptation as follows:

$$K = \tau_{a_+} A_+ - \tau_{a_-} A_- + \tau_x X - \tau_d D,$$

and the total revenue as

$$R(\mathbf{b}^i) = \sum_{t=1}^T b_t^i q_t^a + A + X + D - K.$$

No adaptation costs imply the null hypothesis that $K = 0$. As a first step, this specification is useful because the lack of adaptation costs will be revealed by the data if the estimated coefficients are zero. If they are not, however, then this will indicate the presence of adaptation costs, the exact form of which can then be measured with more scrutiny. (In our empirical analysis the simple linear specification seems to best fit the data.)

To complete the specification of profits, we add a component that captures the loss from submitting irregular bids that are highly skewed. Given our risk neutrality assumption, if a bidder observes a difference between $\mathbf{q}^a$ and $\mathbf{q}^e$ then his incentive is to bid zero on items that are overestimated and a high price on items that are underestimated. As discussed in Section II, however, contractors who submit bids that are too skewed risk having their bids rejected. Hence, skewing bids implies an expected cost on the bidders.

We impose a reduced form penalty that is increasing in the skewness of the bid. Clearly, the degree of skewness will depend on what "reasonable prices" would be. In practice, Caltrans engineers collect information from past bids and market prices to create an estimate $\bar{b}_t$ for the unit cost of contract item t . Thus, given a vector

⁹ Another indirect source of adaptation costs are other extra resources spent due to changes. Jobs that are scheduled to start after the completion of the current job will incur higher costs due to overtime of employees or the hiring of a larger workforce to make up for the extra work.

of prices $\mathbf{b}^i$, a natural measure of skewness would be the distance from the “Blue Book” prices $\bar{\mathbf{b}} = (\bar{b}_1, \dots, \bar{b}_T)$.

Let $P(\mathbf{b}^i | \bar{\mathbf{b}})$ denote the continuously differentiable penalty function of skewing bids satisfying the following assumptions: First, $P(\bar{\mathbf{b}} | \bar{\mathbf{b}}) = 0$ (no penalty from submitting a bid that matches Blue Book prices). Second, $\left. \frac{\partial P(\mathbf{b}^i | \bar{\mathbf{b}})}{\partial b_t^i} \right|_{b_t^i = \bar{b}_t} = 0$ (when the bids match the engineer’s estimates, the first order costs of skewing are zero). These two assumptions seem natural given the practices of Caltrans. Third, $P(\mathbf{b}^i | \bar{\mathbf{b}})$ is strictly convex, and finally, $\lim_{b_t^i \rightarrow 0} \left. \frac{\partial P(\mathbf{b}^i | \bar{\mathbf{b}})}{\partial b_t^i} \right| = \infty$. These last two assumptions guarantee an interior solution to the bidders’ optimization problem in the choice of $\mathbf{b}^i$. For convenience we henceforth drop $\bar{\mathbf{b}}$ and use $P(\mathbf{b}^i)$. This completes the specification of revenues as

$$(1) \quad R(\mathbf{b}^i) = \sum_{t=1}^T b_t^i q_t^a + A + X + D - K - P(\mathbf{b}^i).$$

C. Equilibrium Bidding Behavior

We use Bayesian Nash Equilibrium of the static first-price sealed-bid auction as our solution concept. The game is a scoring auction with independent private values in the spirit of Che (1993) and is a special case of the multidimensional-type model of Asker and Cantillon (2008) where the project is fixed, with the buyer’s objective being trivially fixed given the fixed scoring rule. Similar to Che’s “productive potential” and Asker and Cantillon’s “pseudotype,” our equilibrium behavior will be determined *as if* our bidders have a unidimensional type. The reason is that given the scoring rule, the choice of $s^i = \mathbf{b}^i \cdot \mathbf{q}^e$ is separable from the optimal choice of the actual bid vector $\mathbf{b}^i$.¹⁰ As a result, the Bayesian game will have a unique pure strategy monotonic equilibrium.¹¹

It is therefore useful to decompose bidder i ’s problem into two steps. First, given a score s , we solve for the optimal bid conditional on winning the auction. This will result in the bidding function $\mathbf{b}^i(s)$ (or $b_t^i(s)$, $t = 1, \dots, T$). Then, given $\mathbf{b}^i(s)$, we solve for the optimal score s^i that the bidder would like to submit.

The first problem of choosing the optimal bid function given a score s is given by

$$(2) \quad \begin{aligned} \max_{\mathbf{b}^i(s)} \quad & \sum_{t=1}^T b_t^i q_t^a - \theta^i + A + X + D - K - P(\mathbf{b}^i) \\ \text{s.t.} \quad & \sum_{t=1}^T b_t^i q_t^e = s. \end{aligned}$$

Solving (2) yields $T + 1$ first order conditions (FOCs), the first T being

$$(3) \quad q_t^a - \frac{\partial P(\mathbf{b}^i)}{\partial b_t^i} - \lambda q_t^e = 0 \quad \text{for all } t = 1, \dots, T,$$

¹⁰That is, given the score (price) s , each bidder has an optimal choice of bids *conditional on winning*, $\mathbf{b}^i(s)$, and given this optimal price policy, there is an optimal score $s(\theta^i)$ that is unidimensional.

¹¹This follows from the results of Lebrun (2006).

and the last being the constraint.

After (implicitly) solving for $\mathbf{b}^i(s^i)$, we can complete the bidder's optimization problem of choosing his optimal score s^i . Let $H_j(\cdot)$ be the cumulative distribution function of contractor j 's score, s^j . The probability that contractor i with a score of s^i bids more than contractor j is $H_j(s^i)$. Thus, the bidder's expected profit function is

$$\pi_i(s^i, \theta^i) = [R(\mathbf{b}^i(s^i)) - \theta^i] \times \left[\prod_{j \neq i} (1 - H_j(s^i)) \right].$$

Substituting revenues with (1) and recalling that $\theta_i = \sum_{t=1}^T c_t^i q_t^a$, the contractor's FOC is

$$(4) \quad \sum_{t=1}^T [b_t^i(s^i) - c_t^i] q_t^a = \frac{\sum_{t=1}^T \frac{db_t^i(s^i)}{ds^i} \left[q_t^a - \frac{\partial P(\mathbf{b}^i)}{\partial b_t^i} \right]}{\sum_{j \neq i} \frac{h_j(s^i)}{1 - H_j(s^i)}} - A - X - D + K + P(\mathbf{b}^i).$$

From our assumptions on the densities of types and on the penalty function $P(\cdot)$, $H_j(\cdot)$ is differentiable with density $h_j(\cdot)$, and the FOCs of the two stages of optimization are necessary and sufficient for describing optimal bidder behavior.

A Bayesian Nash Equilibrium is a collection of bid functions, $\mathbf{b}^i(\cdot)$ and scores s^i that simultaneously satisfy the system (3) and (4) for all bidders $i \in \{1, \dots, m\}$. As stated above, there is a unique monotonic equilibrium in pure strategies, and we will therefore use (4) as the basis for our empirical analysis.

The FOC (4) provides some insight into a firm's optimal bidding strategy, and relates to the established literature of bidding without adaptation costs and changes. When $\mathbf{q}^e = \mathbf{q}^a$ and when there are no ex post changes, the FOC (4) reduces to

$$(5) \quad s^i - \mathbf{c}^i \cdot \mathbf{q}^e = \left(\sum_{j \neq i} \frac{h_j(s^i)}{1 - H_j(s^i)} \right)^{-1},$$

which is the FOC for standard first price, private value asymmetric auction models. Our model is therefore a variant of the standard models of bidding for procurement contracts. (See, e.g., Guerre, Perrigne, and Vuong 2000; and Athey and Haile 2006.) That is, the markups reflect the contractors' cost advantage and information rents as captured in the right-hand side of (5).

The innovation in (4) is the introduction of empirically measurable terms that were ignored in previous procurement studies, most notably the adaptation costs reflected in K . To see this, suppose that the contractor expects a deduction of \$1,000. The first order condition suggests that the contractor will raise his bid by $(1 + \tau_d) \times 1,000$. Hence, the total costs of the deductions, as borne by the firm, are indirectly borne by Caltrans.

Clearly, this model abstracts away from what are known to be fundamentally hard problems such as substituting the perfect foresight assumption on changes and actual quantities with a common values specification in which each bidder has signals of these variables. Despite these limitations, however, our first order conditions at a minimum generalize models previously imposed in both the theoretical

TABLE 1—IDENTITIES OF TOP 20 FIRMS

Firm ID	Firm name	Market share	Firm ID	Firm name	Market share
104	Granite Construction Company	25.1%	23	Baldwin Contracting Co.	1.9%
75	E L Yeager Construction Co.	10.6%	410	Brosamer-Granite Joint Venture	1.5%
135	Kiewit Pacific Co.	5.5%	237	Sully Miller Contracting	1.4%
244	Teichert Construction	3.9%	265	West Coast Bridge Inc.	1.2%
12	All American Asphalt	3.3%	186	Pavex Construction	1.2%
262	W Jaxon Baker Inc.	3.3%	234	Steve Manning Construction	1.1%
125	J F Shea Co Inc.	2.6%	162	Mercer Fraser Company	1.1%
147	M C M Construction Inc.	2.4%	126	J McLoughlin Engineering Co.	1.1%
251	Tullis Inc.	2.2%	25	Banshee Construction Co.	1.0%
107	Griffith Company	2.0%	141	Lee's Paving	1.0%
TOTAL					73.4%

Notes: There were a total of 347 active bidders for asphalt concrete construction contracts in our sample between 1999 and 2005. The firms listed above are the top 20 firms from our sample, ranked according to their market share, i.e., the share of total contract dollars awarded. The remaining firms each have less than 1 percent market share and are designated as fringe firms.

and empirical literature, which implicitly impose the assumption that $\tau_l = 0$ for all $l \in \{A_+, A_-, X, D\}$. As we demonstrate shortly, this null hypothesis is strongly rejected by the data.

III. Data

Our unit of observation is a paving contract procured by Caltrans from 1999 through 2005.¹² We index the projects by $n = 1, \dots, N$. Many of the variables in the theoretical section are directly measured in our data, and we use superscript (n) to index these variables for project n . For instance, $b_t^{i,(n)}$ denotes the unit price for item t submitted by bidder i on project n . The sample includes $N = 819$ projects with a total awarded value of \$2.21 billion. There were a total of 3,661 bids submitted by 349 general contractors.

Table 1 lists the top 20 contractors in our dataset and their market share.¹³ Over half of the participating contractors, 193 firms, never won a government asphalt contract during the period and only 2 firms participated in more than 10 percent of the auctions. To account for asymmetry in size and experience, we let $FRINGE_i$ be a dummy variable equal to one if firm i is a “fringe” firm, defined as a firm that won less than 1 percent of the value of contracts awarded. Table 2 compares bidding by the top and fringe firms.

For each project, we collected information from the publicly available bid summaries and final payment forms that include the project number, the bidding date,

¹²Contract details from 2001 and the first half of 2003 are no longer accessible from Caltrans, so our sample does not include contracts from these two periods.

¹³Market size is defined as the value of the winning bids for the projects in our dataset (not the final payments made to the contractors). We focus on contracts for which asphalt is at least one third of the project's monetary value. We exclude contracts that were not awarded to the lowest bidder (which represent only 3.1 percent of all projects). We also exclude 31 contracts for which there was only one recorded bidder and 65 contracts for which there is no itemized record of the final payment or pages missing from record files. Lastly, during this time period, there were 22 paving contracts that were structured as “A+B” contracts, where bidders submit a bid on the number of days to completion as well as unit bids on itemized tasks. These types of contracts only appear later in the sample and are excluded. Table A1 in the online Appendix shows the bidding activities of the top 20 firms.

TABLE 2—COMPARISON BETWEEN FRINGE FIRMS AND FIRMS WITH OVER 1 PERCENT MARKET SHARE

	Fringe firms	Nonfringe firms
Number of firms	327	20
Number of wins	388	431
Number of bids submitted	2,269	1,392
Average wins per firm	1.2	21.6
Average bid submitted*	\$3,244,907	\$6,175,811
Average distance to job site (miles)*	99.13	109.64
Average capacity*	\$2,250,619	\$30,099,627
Average backlog at time of bid*	\$94,322	\$6,417,988

Note: *The above averages were calculated by first calculating the average for each bidder, then averaging these means over the fringe and non-fringe firms, respectively.

the location of the job, other information about the nature of the job, and bidder identities with their itemized bids. Projects have an average of 33 items, although one project has 326 items. For each item, we have the unit prices for all bidders, along with the estimated quantity.

We also obtained the engineer's estimate of the project's cost. This measure, provided to potential bidders before proposals are submitted, is intended to represent the "fair and reasonable price" the government expects to pay for the work to be performed. This estimate can be thought of as $\sum_{t=1}^T [\bar{b}_t q_t^{e(n)}]$, the dot product of Blue Book prices and the estimated quantities for project n . Caltrans measures $\bar{b}_t$ using the Blue Book prices published in the Contract Cost Data Book (CCDB), an item-level data summary prepared annually by Caltrans' Division of Office Engineer.¹⁴ We have merged this information into our dataset. Thus, a unique feature of our data is that we directly measure all the tasks and we have a cost estimate for every task. Such detailed cost information is rare in procurement studies and it allows us to incorporate an appealing set of controls in our regression analysis.

From the final payment forms, we collect data on the actual quantities, $q_t^{a(n)}$, used for each item. Additionally, the forms record the adjustments, extra work, and deductions that contribute to the total price of the project. These correspond to the variables $A_+^{(n)}$, $A_-^{(n)}$, $D^{(n)}$, and $X^{(n)}$ introduced in the previous section.

To account for the advantage of geographic proximity (transportation cost) we measure the distance of firm i from project n , denoted as $DIST_i^{(n)}$.¹⁵ As expected, contractors who submit the lowest bids also tend to have the shortest travel distances, reflecting their cost advantage.

¹⁴ See the "Plans, Specifications, and Estimates Guide," published by the Caltrans' Division of Office Engineer for information about the formation of this estimate. Not all items have values in this source. For items with no CCDB value, we derive an estimate for $\bar{b}_t$ using the average of the low bidder's unit price on an item over all contracts in a given district and year. This averaging method is consistent with the method professional estimating companies use to create benchmark prices for the CCDB. This $\bar{b}_t$ had an R^2 of 0.66 when regressed on the estimates we received from the CCDB. Furthermore, when we regress a constructed measure of $\sum_t \bar{b}_t q_t^{e(n)}$ on the engineer's total estimate from our data, the R^2 is 0.985.

¹⁵ The contract often provides the location of the project as the cross streets at which highway construction begins and ends. Where the information is less precise we use the city's centroid or a best estimate based on the post mile markers and highway names included in the contract. We record distance using the address of each bidder as calculated by Mapquest. When a bidder has multiple office locations we use the one closest to the job site. For projects with multiple locations we take the average of the distances to each location. Using Mapquest's estimated travel time rather than distance produced quantitatively similar results.

TABLE 3—SUMMARY STATISTICS

	Observations	Mean	SD	Min.	Max.
<i>Across contracts under consideration</i>					
Winning bid	819	2,697,385	6,859,444	51,625.50	99,599,234
Markup: (winning bid-estimate)/estimate	819	-0.0545	0.1982	-0.6166	1.188
Normalized bid: winning bid/estimate	819	0.9455	0.1982	0.3834	2.188
Second lowest bid	819	2,878,726	7,273,552	60,764.50	106,988,376
Money on the table: second bid-first bid	819	181,340.70	474,494.5	67.5	7,389,144
Normalized money on the table: (Second bid-first bid)/estimate	819	0.0763	0.0756	0.0002	0.7486
Number of bidders	819	4.47	2.15	2	19
Distance of the winning bidder	819	82.37	114.5	0.1	1,066
Utilization rate of the winning bidder	819	0.1068	0.2023	0	1
Distance of the second lowest bidder	819	95.74	119.65	0.1	996
Utilization rate of the second lowest bidder	819	0.1144	0.2175	0	1
<i>Across bids submitted</i>					
Normalized bid	3,661	1.044	0.2362	0.3834	3.1051
Distance to job site	3,661	94.19	129.9	0.1	2,857
Backlog at time of bid	3,661	4,656,086	14,274,630	0	150,411,535
Capacity	3,661	29,793,789	50,680,540	0	150,411,535
Utilization (backlog/capacity)	3,661	0.1029	0.2193	0	1
Minimal distance among rivals	3,661	38.64	53.55	0.1	618.62
Minimal utilization among rivals	3,661	0.0168	0.075	0	1

A firm's bidding behavior may be influenced by its production capacity and project backlog. Firms that are working close to capacity face a higher opportunity cost when considering an additional job. Following Porter and Zona (1993), we construct a measure of backlog from the record of winning bids, bidding dates, and project working days. We assume that work proceeds at a constant pace over the length of the project, and define the variable $BACKLOG_i^{(n)}$ to be the remaining dollar value of projects won but not yet completed at the time a new bid is submitted.¹⁶ We then define $CAPACITY_i^{(n)}$ as the maximum backlog experienced for any day during the sample period, and the utilization rate $UTIL_i^{(n)}$ as the ratio of backlog to capacity. For those firms that never won a contract, the backlog, capacity, and utilization rate are all set to 0.¹⁷

Firms may take into account their competitors' positions when devising their own bids. We therefore include measures of their closest rival's distance and utilization rate. We define $RDIST_i^{(n)}$ as the minimum distance to the job site among i 's rival bidders on project n . Likewise, $RUTIL_i^{(n)}$ is the minimum utilization rate among i 's rival bidders on project n .

Summary statistics for the projects and the bids are provided in Tables 3 and 4 (as well as in Tables A2 and A3 in the online Appendix). There is noticeable heterogeneity in the size of projects awarded: the mean value of the winning bid is \$2.7 million with a standard deviation of \$6.9 million. The difference between the

¹⁶This measure was constructed using the entire set of asphalt concrete contracts, even though a few of these were excluded from the analysis. Since we lack information from the previous year, the calculated backlog will underestimate the true activity of firms during the first few months of 1999.

¹⁷Bajari and Ye (2003) show that the opportunity cost of capacity enters into the FOCs like a deterministic cost shifter. This assumption is valid if bidders are indifferent about which of their competitors wins a project so that there is no incentive to strategically manipulate the capacities of competitors. See Jofre-Bonet and Pesendorfer (2003) for a dynamic analysis of capacity constrained bidders.

TABLE 4—IMPORTANCE OF EX POST CHANGES

	Observations	Mean	SD	Min.	Max.
Adjustments	819	142,035	832,908	−195,727	15,450,334
Adjustments/estimate	819	0.021	0.0489	−0.2172	0.3962
Extra work	819	176,256	657,249	0	14,697,661
Extra work/estimate	819	0.0608	0.0829	0	0.8455
Deductions	819	−8,615	94,642	−2,530,053	0
Deduction/estimate	819	−0.0021	0.0095	−0.1928	0
CCDB overrun = (ActQ − EstQ) × CCDB price	819	−62,204	486,982	−9,462,806	1,699,937
CCDB overrun/estimate	819	−0.0222	0.2366	−6.34	0.2859
Final payment − winning bid	819	190,376	1,436,883	−24,111,355	21,190,429
(Final payment − winning bid)/estimate	819	0.0577	0.1187	−0.6591	0.653

Notes: The CCDB Overrun is meant to reflect the dollar overrun due to quantities that were misestimated during the procurement process. It is only a partial measure of the quantity-related overrun, since some of the nonstandard contract items do not have a corresponding price estimate from the Contract Cost Data Book (CCDB). The engineer's estimate was used to normalize this and the other measures.

TABLE 5—SKEWED BIDDING REGRESSIONS

Variable	OLS	Item code fixed effects
Percent unit overrun	0.0465 (3.61)	0.0535 (3.94)
Constant	1.7829 (55.99)	1.7826 (56.97)
R^2	0.00	0.04
Observations	109,624	109,624

Notes: The dependent variable is the unit price bid on each contract item, normalized by the Contract Cost Data Book (CCDB) value. The percent unit overrun is the percent difference between the actual and estimated quantities reported for that item. Because the pricing strategy a bidder uses for one item is related to its pricing strategy for all other items within a contract, we expect the residuals to be correlated across items for a given bidder and contract. To account for this correlation, standard errors clustered by bidder-contract are used to compute *t*-statistics, shown in parentheses.

first and second lowest bids averages \$181,340, meaning that bidders leave some “money on the table.” On average, the projects require 108 working days to complete, and several change orders are processed. The final price paid for the work exceeds the winning bid by an average of \$190,376 (5.8 percent of the estimate). As Table 4 shows, some of this discrepancy can be attributed to over and underruns on project items. Large deviations also induce a correction to the item's total price, captured by the value of adjustments. In our sample, the mean adjustment is \$142,035. Compensation for extra work negotiated after change orders, as well as deductions, contribute to the difference, averaging \$176,256 and (−\$8,615) respectively. These suggest a substantial degree of incompleteness in the original contracts.

As Athey and Levin (2001) show, contractors can raise profits by skewing their bids upwards (downwards) on items that are expected to overrun (underrun). Table 5 examines bid-skewing by reporting a regression of the unit prices on the percent by which that particular item overran. The left-hand-side variable is the unit price divided by an estimate of the CCDB unit costs.¹⁸ The coefficient on percent overrun is 0.0465, which is statistically significant at the 5 percent level. That is, if a

¹⁸The CCDB only contains estimates for the more common variations of the construction items so some items are not reported. Therefore, this regression only uses 92 percent of all the item-unit bids submitted.

contractor expected a 10 percent overrun on some item, he would shade his bid up on that item by approximately 0.5 percent, a modest amount. When we allow for heteroskedasticity within an item code by using fixed or random item effects, the coefficient on percent overrun is similar, although with 1,519 types of items, these effects do not add much explanatory power to the regression. This evidence suggests that incentives to skew are not a major determinant of the observed bids.

IV. Reduced Form Estimates

A. Bid Regressions

We begin our analysis by performing some common reduced form regressions to determine what best explains the total bids. A typical reduced form approximation to equation (5) implies that $\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)}$ should be determined by costs and measures of market power.

We control for firm i 's costs using four terms. First is the engineer's cost estimate. A regression of $\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)}$ on the engineer's cost estimate, $\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{e,(n)}$, yields an R^2 of 0.982 and a coefficient equal to 1.039, making the engineer's cost estimate an excellent cost control. Second, a firm's own distance to the project, $DIST_i^{(n)}$ will influence transportation costs. Third, $UTIL_i^{(n)}$ will measure firm i 's free capacity. Finally, we allow the bids to differ by firm size and include an indicator, $FRINGE_i^{(n)}$ for fringe firms as described above.

A bidder's markup over costs will also depend on publicly observed information about its competitors, for which we use three terms as follows. Firm i will have more market power when (i) the distance of its closest competing rival to the project, $RDIST_i^{(n)}$, is farther, (ii) when its rivals capacity utilization, $RUTIL_i^{(n)}$, is higher, and (iii) when the number of bidders, $NBIDS^{(n)}$, is greater.

The impact of our covariates on the bids will be proportional to the engineer's cost estimate ($\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{e,(n)}$). One would expect increasing i 's distance by ten miles will raise the bid more on a contract with a \$5 million dollar estimate than one with a \$500,000 dollar estimate. It is also natural to expect the variance of the error term to be proportional to ($\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{e,(n)}$). The efficiency of our regression estimates would be improved by dividing our regression through by ($\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{e,(n)}$) to control for heteroskedasticity. Therefore, we propose estimating the following equation:

$$(6) \quad \frac{\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)}}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{e,(n)}} = \alpha_0 + \alpha_n + \alpha_i + \alpha_2 DIST_i^{(n)} + \alpha_3 UTIL_i^{(n)} + \alpha_4 FRINGE_i^{(n)} \\ + \alpha_5 RDIST_i^{(n)} + \alpha_6 RUTIL_i^{(n)} + \alpha_7 NBIDS^{(n)} + \varepsilon_i^{(n)}.$$

We include project fixed effects (α_n) and firm fixed effects (α_i). Project fixed effects control for information that is publicly observed by all the firms but not by us. Firm fixed effects control for omitted cost shifters of firms that are persistent across auctions. Regressions such as (6) are often used. (See, e.g., Porter and Zona 1993.)

Results from estimating (6), and its variants, are displayed in Table 6. Fringe status is significant and has positive signs as expected; fringe firms bid 3.4 to 4.7 percent higher than more established competitors. Distance is positive and

TABLE 6—STANDARD BID FUNCTION REGRESSIONS

Variable	(1)	(2)	(3)	(4)	(5)
<i>DIST</i>	0.0067 (2.13)	0.0093 (2.76)	0.0034 (1.04)	0.0076 (3.38)	0.0091 (3.94)
<i>RDIST</i>	0.0332 (3.77)	0.0217 (2.44)	0.0232 (2.70)	−0.0054 (−0.61)	0.0047 (0.58)
<i>UTIL</i>		0.0149 (0.80)	0.0429 (2.50)	0.0126 (0.81)	0.0074 (0.50)
<i>RUTIL</i>		−0.1377 (−2.66)	−0.1299 (−2.14)	0.0334 (0.71)	−0.0332 (−0.77)
<i>FRINGE</i>		0.0483 (5.76)		0.0379 (7.27)	0.0403 (7.88)
Number of bidders		−0.015 (−10.30)	−0.0171 (−10.33)		−0.016 (−5.45)
Constant	1.0253 (187.95)	1.0795 (92.52)	1.124 (110.34)	1.014 (165.79)	1.0966 (61.48)
Fixed/random effects	No	No	Firm FE	Project FE	Project RE
R^2	0.008	0.035	0.216	0.733	0.035
Observations	3,661	3,661	3,661	3,661	3,661

Notes: The dependent variable is the total bid divided by the engineer's estimate, where the total bid is the dot product of the estimated quantities and unit prices. Distances are measured in 100 miles. Cluster-robust standard errors are used to compute t -statistics, shown in parentheses.

significant in all but one specification; a firm located 94 miles from the project (the sample average) would bid about 1 percent more than a firm that is adjacent to the project. Increases in rival's distance allows a firm to increase its own bid by 2–3 percent for every 100 miles. The number of firms in a market has the expected sign; adding an additional bidder to the job lowers bids by about 1.5 percent.

The goodness of fit in columns 1 and 2 is low. In columns 3 and 4, we add project and firm fixed effects. The results suggest that both of these variables add considerably to goodness of fit, particularly project fixed effects. These effects capture characteristics such as the condition of the job site, the difficulty of the tasks, and anticipated changes. As column 5 shows, substituting project fixed effects with project random-effects has almost no impact on the results.

B. Accounting for Changes and Adaptation Costs

While regressions such as those in Table 6 are common, equation (4) suggests that they suffer from two sources of misspecification. First, the dependent variable is the total estimated bid, $\mathbf{b}^i \cdot \mathbf{q}^e$, instead of the (expected) total payment, $\mathbf{b}^i \cdot \mathbf{q}^a$. Second, the regressions ignore the anticipated changes to payments due to adjustments, extras, and deductions. Based on equation (4), we respecify the reduced form regression as follows:

$$\begin{aligned}
 (7) \quad \frac{\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{a,(n)}}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}} = & \alpha_n + \alpha_i + \alpha_2 DIST_i^{(n)} + \alpha_3 UTIL_i^{(n)} + \alpha_4 FRINGE_i^{(n)} \\
 & + \alpha_5 RDIST_i^{(n)} + \alpha_6 RUTIL_i^{(n)} + \alpha_7 NBIDS^{(n)} + \varepsilon_i^{(n)},
 \end{aligned}$$

where

$$\alpha_n = \beta_1 + \beta_2 A_+^{(n)} + \beta_3 A_-^{(n)} + \beta_4 X^{(n)} + \beta_5 D^{(n)} + \varepsilon_n.$$

The regression in (7) is similar to that in (6). However, the dependent variable is now consistent with (4). As before, we correct for heteroskedasticity related to project size by dividing through by an estimate of that size, $\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}$.

Equation (4) implies that the marginal impact of an extra dollar of change identifies the adaptation costs in our model. Following the logic of (4), our estimates of β_2 through β_5 offer estimates of the adaptation cost coefficients as follows:¹⁹

$$\begin{aligned} \beta_2 &\equiv -(1 - \tau_{a+}) & \beta_4 &\equiv -(1 - \tau_x) \\ \beta_3 &\equiv -(1 + \tau_{a-}) & \beta_5 &\equiv -(1 + \tau_d). \end{aligned}$$

The results of these least-squares regressions appear in Table 7. As columns 1 and 2 demonstrate, including only the cost shifters of the firm and its competitors as covariates yields results similar to columns 2 and 4 in Table 6.²⁰ Project fixed effects absorb a great deal of variation, consistent with unobserved project-specific heterogeneity. Note that the regression in Table 7, column 2, is almost identical to the regression in Table 6, column 4, yet the R^2 increases slightly when the dependent variable is unit prices times the *actual quantities*, as suggested by (4). This is subtle evidence that ex post information better explains the observed bids.

Next, we include the ex post changes: positive and negative adjustments ($PosADJ^{(n)}$ and $NegADJ^{(n)}$), extra work ($EX^{(n)}$), and deductions ($DED^{(n)}$), all normalized by dividing through by $\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}$. The regression results without project-specific effects are shown in column 3, with project fixed effects in column 4, and with project random effects in column 5.²¹ The coefficients on ex post changes are similar in all three columns, and most are highly significant.²² However, the coefficient on deductions is not statistically significant at standard levels. The lack of significance is most likely a result of the small variation in this type of ex post change. Deductions only affect 209 of the 819 contracts (compared to 752 contracts with extra work and 536 contracts with positive or negative price adjustments), and much

¹⁹The coefficient on X in (4) is $-(1 + \tau_x)$, and because (7) regresses the bid on covariates instead of the cost, the signs are reversed and $\beta_4 = -(1 - \tau_x)$. A similar logic applies to the other ex post coefficients.

²⁰The coefficient on $NBIDS^{(n)}$ is not always negative, and is never significant. This is caused by 5 contracts with more than 13 bidders, all of which run over by an average of 7 percent. Excluding them gives the expected negative sign.

²¹In column 4, we estimate the project fixed effects along with the firm-specific controls in a first step, then regress those fixed effects on measures of ex post changes to the project. We acknowledge that, because the number of bids per contract is small and not approaching infinity, these fixed effects estimates are not consistently estimated. We prefer the results using random effects, but include both to show how little the estimates change under different specifications.

²²We also report the impact of changes to individual item quantities on bids. Our model predicts that this could impact the bids through the skewing penalty. We constructed a measure, $NOverrun^{(n)}$, which is defined as the sum of the dollar overrun on individual items, divided by the project estimate. This dollar overrun is computed by multiplying the difference in the actual and estimated quantity by the item cost estimate reported in the Contract Cost Data Book, $\bar{b}_i$. Since not all contract items are contained in the data book, $NOverrun^{(n)}$ should be thought of as a partial project overrun due to quantity changes in the more standard items. The coefficient on $NOverrun^{(n)}$ is significant, but very small in magnitude.

TABLE 7—BID FUNCTION REGRESSIONS USING ACTUAL QUANTITIES INSTEAD OF ESTIMATES

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>DIST</i>	0.0227 (5.91)	0.0089 (3.59)	0.0221 (5.94)	0.0089 (3.61)	0.0123 (4.98)	0.0217 (5.85)	0.0089 (3.61)	0.0119 (5.32)
<i>RDIST</i>	0.0347 (3.30)	−0.0015 (−0.17)	0.0353 (3.41)	−0.0013 (−0.15)	0.0131 (1.83)	0.0347 (3.24)	−0.0013 (−0.15)	0.0116 (1.57)
<i>UTIL</i>	0.0401 (1.85)	0.0180 (1.02)						
<i>RUTIL</i>	0.0972 (1.50)	0.0364 (0.80)						
<i>FRINGE</i>	0.002 (0.21)	0.0355 (6.34)	−0.00004 (−0.00)	0.034 (6.46)	0.0293 (5.59)	0.0001 (0.02)	0.034 (6.46)	0.0297 (5.32)
Number of bidders	0.0055 (1.05)		0.0058 (1.11)	0.0024 (0.47)	−0.0032 (−0.80)	0.0062 (1.21)	0.003 (0.58)	−0.0026 (−0.73)
<i>NPosAdj</i>			0.8032 (5.89)	0.819 (5.85)	0.7712 (6.39)	0.8758 (3.87)	0.9044 (3.82)	0.9194 (4.67)
<i>NNegAdj</i>			−1.7988 (−2.23)	−1.7367 (−2.24)	−1.8894 (−2.25)	−1.6894 (−1.78)	−1.5647 (−1.72)	−1.5473 (−1.62)
<i>NEX</i>			0.1644 (1.74)	0.1647 (1.77)	0.1559 (1.79)	0.2234 (1.64)	0.2261 (1.66)	0.2089 (2.49)
<i>NDED</i>			−1.0231 (−1.31)	−1.3268 (−1.84)	−0.958 (−1.46)	−1.746 (−0.99)	−2.4838 (−1.30)	−2.2187 (−1.58)
<i>NOverrun</i>			0.0057 (5.46)	0.0059 (5.46)	0.0054 (5.76)	0.0065 (3.72)	0.0068 (3.70)	0.0070 (3.94)
Constant	0.9375 (32.68)	0.9771 (152.49)	0.9054 (31.74)	−0.0518 (−1.91)	0.9556 (44.80)	0.8967 (29.73)	−0.0628 (−2.17)	0.9443 (44.59)
Fixed/random effects	No	Project FE	No	Project FE	Project RE	No	Project FE	Project RE
Instruments	None	None	None	None	None	Resident Engineer	Resident Engineer	Resident Engineer
R^2	0.0254	0.7623	0.0738	0.7621	0.0599	0.0712	0.7621	0.0577
Observations	3,661	3,661	3,661	3,661	3,661	3,661	3,661	3,661

Notes: The dependent variable is the vector product of the unit price bids and the actual quantities, divided by a measure of the project size ($\bar{b} \cdot q^{act}$). Cluster-robust standard errors are used to compute *t*-statistics, shown in parentheses. *NOverrun* is a measure of the quantity-related overrun on standard contract items (those that have a CCDB unit price estimate). This overrun is calculated as the vector product of the CCDB prices (where available) and the difference between actual and estimated quantities.

of the variance is driven by a single contract with a particularly large deduction of almost \$2.5 million.²³

The results provide evidence that adaptations impose significant frictions. For example, in columns 3–5 of Table 7, the coefficient of about (−1.8) on negative adjustments suggests that negotiations over negative adjustments carry with them \$0.8 adaptation costs for every dollar in adjustments. If bidders anticipate negative adjustments then they tend to raise their bids not only to recoup the expected loss, but also to recover the adaptation costs they will expend while haggling over price changes.

²³ This contract was assessed \$2.38 million in liquidated damages for completing the project 119 days late, with the other deductions coming from quality and compliance penalties. Dropping this one contract does not substantially affect the results.

With no adaptation costs the coefficients on positive adjustments should equal -1 . The coefficient of about 0.8 implies that firms actually *raise* their bids when they expect this additional ex post compensation because they expect to spend \$1.8 in adaptation costs for every dollar they obtain in adjustment compensation. This may seem bizarre, but a simple rent-seeking story can rationalize this result.²⁴

Similarly, the coefficient of 0.16 on extra work implies that firms expect to spend \$1.16 in costs for every dollar they obtain in adjustment compensation.²⁵ Adaptation costs on deductions are much smaller and, as mentioned above, less significant.

C. Endogenous Ex Post Changes and Omitted Costs

A concern with the analysis above is that ex post changes may be correlated with the error term because of omitted costs that are observed by the bidders for which we cannot control. For instance, a project in a more mountainous area will impose higher production costs and will be more likely to require changes due to the more challenging terrain. If so, projects with more changes have higher costs not because of adaptation costs but because of higher production costs in rough terrains. Another plausible story is that very complex projects impose serious delays and difficulties that increase the labor costs of production. If these delays are the source of adjustments and deductions then the increased bids may actually be a consequence of the increased production costs.

Recall that we observe the actual quantities used for each itemized component of the contract, and have excellent cost data from the CCDB. If delays are accompanied by higher costs of production, much of this will be captured by higher actual quantities of specified contract items because these are controlled for by $\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}$, which in a simple univariate regression explains 97.4 percent of the variation in the ex post bid. Hence, it seems safe to believe that any omitted costs will be negligible. Nonetheless, we proceed to implement an instrumental variable approach to correct for the possible endogeneity of ex post changes to compensation by identifying an exogenous variable that affects ex post changes but does not affect the unobserved project-specific production costs. In particular, we use the identity of the Caltrans project engineer who supervised the project (i.e., the project manager) to capture exogenous shocks to ex post changes.

While Caltrans highway contracts have numerous clauses devoted to changes, incomplete contracts imply that project engineers have considerable discretion over the scope of changes and deductions, and the process through which these changes are governed. It is well known in the industry that there is considerable heterogeneity in the propensity of engineers to make changes to the contract or impose deductions. Like in many fields, some engineers are naturally adept at dealing with conflict and solving disputes while others are not. Some engineers are harder to work with because of their propensity to impose deductions and adjustments,

²⁴ For example, imagine that ex post Caltrans wishes to impose negative adjustments of $-\$1$, which will be enforced if the firm does not contest. Instead, the firm can pay haggling costs (time, legal fees, etc.) of \$1.8, after which it will be able to collect a positive adjustment of \$1. Since the costs are lower than the benefit (\$2), this will result in an observation of \$1 in positive adjustments ex post, but a rational increase in the ex ante bid of \$1.8.

²⁵ Our regressions imperfectly control for costs from extra work. In our structural model, presented in the next section, we attempt to deal with this by bounding the potential bias from omitted costs.

causing disruptions to the efficient flow of work and imposing undesirable renegotiation costs. Because every project is assigned to an engineer who handles several projects, the identity of the engineer will shift the distribution of ex post changes to the contracts he manages, *independent* of the specific project characteristics.

For the engineer's identity to be a valid instrument it must be: (i) correlated with the endogenous variables (changes), and (ii) uncorrelated with the error term. Condition (i) is fairly easy to verify. Our 819 contracts were handled by 334 unique engineers. After dropping the 154 engineers who handle only one contract we are left with 180 engineers, each managing at least 2 of the remaining 665 contracts.²⁶ First-stage regressions of the potentially endogenous ex post changes on dummy variables for these engineers confirm instrument strength particularly for positive adjustments to compensation and overruns on itemized contract elements, but suggest that they may be weak with regard to extra work and deductions. See the online Appendix for *F*-statistics and further details about instrument quality.

Condition (ii) is not possible to verify directly since it is an identifying assumption, but it is very plausible because of the way that contracts are assigned to engineers according to the following sequence of events.²⁷ The Caltrans engineering staff first draws the plans and specifications for a given project. The project is then publicly advertised and the plans, specifications, and other bidding documents are made available to bidders. The location of the project allows bidders to determine the district office from which the engineer will be assigned. There are a handful of engineers at a given district office and they are matched to projects based upon their expertise and availability. In most cases the engineer is assigned early and noted on the project plans for bidders to contact prior to bidding with questions about certain specifications. After bids are submitted and a winner is chosen, work begins and changes to the project are made based upon work progress and site conditions.

Identification also requires that our instrument be mean-independent of the unobserved costs $\varepsilon^{(n)}$. One might worry that project engineers predisposed to change the contract are assigned in a nonrandom way to more or less complicated projects. We find, instead, that the best predictor of the assignment of a project engineer to a contract is which of the 12 district offices the engineer works at: 96 percent of the engineers work in a single district. However, district dummies alone do not predict $A_+^{(n)}$, $A_-^{(n)}$, $X^{(n)}$, and $D^{(n)}$. All districts have, on average, a similar share of projects that experience large changes. The scarce supply of engineers in any given district, each with a limited capacity to take on projects, generates exogeneity in how engineers are assigned to projects with many changes.

Another test of random assignment is to regress measures of engineer experience on ex post changes. We observe how many projects are assigned to a particular engineer, which we interpret as a proxy for experience or productivity. We regress this variable on $A_+^{(n)}$, $A_-^{(n)}$, $X^{(n)}$, and $D^{(n)}$. Nonrandom assignment implies that more experienced engineers are assigned to projects with more changes. However, the R^2

²⁶ In this set, the mean number of contracts per engineer is 3.69 with a standard deviation of 2.04. The total value of projects managed by each engineer has a mean of \$9.94 million and a standard deviation of \$13.6 million (using the engineer's estimate). Our data consist of only asphalt contracts, while Caltrans engineers also manage other contracts, hence the small number of contracts per engineer.

²⁷ Any variable known at the time of bidding is a valid instrument (Hansen and Singleton 1982) because anything known at the time of bidding cannot be correlated with the forecast error of payoff relevant variables.

in this regression was less than 0.01 and none of the coefficients on ex post change variables were significant.

We present the instrumental variable estimates in columns 6, 7, and 8 of Table 7. In column 7, as in column 4, we regress the project level fixed effects on measures of ex post changes to the project. Notice that the estimated values using our instruments are similar to those from the OLS estimation in columns 3 through 5. The estimated adaptation costs for adjustments are very similar in magnitude and significance, while those for extra work and especially deductions are higher in magnitude. All are statistically significant in the fixed effects regression, and most are in the other two specifications (columns 6 and 8).

V. Structural Estimation

The reduced form analysis suggests that contractors build sizeable adaptation costs into their bids in anticipation of ex post changes. However, much of the literature on contracting has focused on other sources of distortion, primarily rents from private information and market power, and, more recently, strategically skewed bidding. In order to assess the relative magnitude of these distortions alongside those coming from adaptation costs, we require a method for structurally estimating the model discussed in Section III.

Our estimation approach builds on the two-step nonparametric estimators discussed in Elyakime et al. (1994) and in Guerre, Perrigne, and Vuong (2000). In the first step, we estimate the density and CDF of the bid distributions for project n , denoted by $h_j^{(n)}(s^i)$ and $H_j^{(n)}(s^i)$ respectively. In the second step, we estimate a particular form of the penalty from skewed bidding and the adjustment cost coefficients, τ_{a+} , τ_{a-} , τ_d , and τ_x , by using the first order conditions in (4) to form a GMM estimator.

This approach also addresses some potential sources of misspecification in the reduced form analysis.²⁸ First, we note that since bidders will be uncertain about the magnitude of ex post changes, the first order conditions should include the expected values of $DED^{(n)}$, $EX^{(n)}$, $PosADJ^{(n)}$, and $NegADJ^{(n)}$ instead of their actual values. The standard econometric analysis of measurement error suggests that this will bias our reduced form estimates of adaptation costs. Second, our reduced form regressions imperfectly approximate the first order conditions. For instance, we attempt to capture market power by including $RDIST_i^{(n)}$ and $RUTIL_i^{(n)}$ as regressors. However, we can more precisely assess market power by directly including the probability of winning, as implied by the first order conditions (e.g., the right-hand-side term in (5)).

Finally, we note that the structural model allows us to better flesh out the interpretation of the error term, which is important for assessing the plausibility of the instruments used for estimation. We propose instruments that allow for consistent estimation of the adaptation costs when there are two sources of endogeneity. The first are the unobserved cost shocks discussed in Section IVC. The second are the expectational errors, mentioned above.

²⁸ We acknowledge that our structural analysis may bring about other misspecifications, but our primary goal is to assess the relative magnitude of sources of distortion.

While the structural model uses different econometric methods, we shall find a great deal of consistency with our reduced form results.

A. Estimating Bid Distributions

Since we wish to include measures of firm-specific distance and other controls for cross firm heterogeneity, nonparametric approaches would suffer from a curse of dimensionality.²⁹ Hence, we will estimate $h_j^{(n)}(s^i)$ and $H_j^{(n)}(s^i)$ semiparametrically. We first run a regression similar to those in Table 6:

$$(8) \quad \frac{\mathbf{b}_j^{(n)} \cdot \mathbf{q}^{e,(n)}}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^e} = x_j^{(n)'} \mu + u^{(n)} + \varepsilon_j^{(n)},$$

where, as before, the dependent variable is the normalized estimated bid and $x_j^{(n)}$ includes the firm's distance and whether or not it is a fringe firm.³⁰ We also include an auction-specific random effect, $u^{(n)}$, to control for project-specific characteristics that are observed by the bidders but not the econometrician.³¹

Let $\hat{\mu}$ denote the estimated value of μ and let $\hat{\varepsilon}_j^{(n)}$ denote the fitted residual. We assume that the residuals to this regression are i.i.d. with distribution $G_{\mathcal{N}}(\cdot)$ where $\mathcal{N}$ indexes the number of bidders in the auction. That is, consistent with the model, we treat all auctions with the same number of bidders in the same way, but account for the effect of the number of bidders on the distribution of bids. The i.i.d. assumption would be satisfied if the noise on total costs had a multiplicative structure, which we describe in detail in the next subsection. Under these assumptions, for project n ,

$$\begin{aligned} H_j^{(n)}(b) &\equiv \Pr \left(\frac{\mathbf{b}_j^{(n)} \cdot \mathbf{q}^{e,(n)}}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{e,(n)}} \leq \frac{b}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{e,(n)}} \right) = \Pr \left(x_j^{(n)'} \mu + u^{(n)} + \varepsilon_j^{(n)} \leq \frac{b}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{e,(n)}} \right) \\ &\equiv G_{\mathcal{N}} \left(\frac{b}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{e,(n)}} - x_j^{(n)'} \mu - u^{(n)} \right). \end{aligned}$$

²⁹ Specifically, the sample size required in order to achieve the same level of precision in the estimates increases dramatically as the number of covariates included in the kernel regression increases. See, for example, Table 4.2 of Silverman (1986), as well as the *Handbook of Econometrics* chapter by Athey and Haile (2007) and the references therein.

³⁰ As shown later in equation (9) we are assuming that a bidder's underlying private information about costs have a multiplicative structure. That is, costs can be written as a product of $\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^e$ and a bidder specific private information shock. It follows from the linearity of expected utility and the definition of Bayes-Nash equilibrium that if we divided all costs by $\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^e$ and all bids by the same constant, this transformed bid function is also an equilibrium. This idea is noted in both Athey and Haile (2007, section 6.1.1) and Haile, Hong, and Shum (2003).

³¹ As Krasnokutskaya (2011) has emphasized, failure to account for this form of unobserved heterogeneity may lead to a considerable bias in the structural estimates. However, the strategies proposed in the literature for dealing with such heterogeneity, including Krasnokutskaya's deconvolution-style estimator and the parametric approach of Athey, Levin, and Seira (2011), are not straightforward to apply in our more complicated framework. Instead, our use of random effects requires that the auction-specific effect $u^{(n)}$ is mean independent of the bidder's private information, a restriction that is analogous to the independence requirements of Krasnokutskaya's more general deconvolution approach. As a robustness check we also estimated a version of the model with fixed effects and found little quantitative change in our results. We do not rely upon these fixed effects results, however, because with a finite number of bidders per contract, we have an incidental parameters problem and cannot consistently estimate the distribution of residuals.

That is, the distribution of the residuals, $\varepsilon_j^{(n)}$ can be used to derive the distribution of the observed bids. We estimate $G_{\mathcal{N}}$ using the distribution of the fitted residuals $\hat{\varepsilon}_j^{(n)}$ for all contracts with the same number of bidders, $\mathcal{N}$, as contract n . Then we recover an estimate of $H_j^{(n)}(b)$ by substituting in this distribution in place of $G_{\mathcal{N}}$. An estimate of $h_j^{(n)}(b)$ can be formed using similar logic. We note that both $H_j^{(n)}(b)$ and $h_j^{(n)}(b)$ will be estimated quite precisely.³² Given the estimates $\hat{H}_j^{(n)}$ and $\hat{h}_j^{(n)}$ we construct an estimate for $\left(\sum_{j \neq i} \frac{\hat{h}_j^{(n)}(\mathbf{b}_i \cdot \mathbf{q}^e)}{1 - \hat{H}_j^{(n)}(\mathbf{b}_i \cdot \mathbf{q}^e)} \right)^{-1}$. Note that this term accounts for varying numbers and identities of bidders across auctions. In particular, for any given auction, let j index the bidders, so that if the number or identity/fringe status of bidders across auctions changes, the set of firms indexed by j will also change.³³

B. Estimating Adaptation Costs

As demonstrated in Section V, the engineering cost estimate, $\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}$, is an excellent predictor of the bids. Therefore, we assume that firm i 's cost is a variant of the engineer's cost estimate with the following multiplicative structure:

$$(9) \quad \theta_i^{(n)} = \mathbf{c}_i^{(n)} \cdot \mathbf{q}^{a,(n)} \equiv \bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)} (1 + \tilde{c}_i^{(n)}).$$

That is, *actual* total costs for firm i are a deviation from the engineer's cost estimate represented as a random variable $\tilde{c}_i^{(n)}$ times the engineering estimate $\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}$. The assumption in (9) is similar to the multiplicative structure used in Krasnokutskaya (2011). A similar assumption is also implicit in Hendricks, Pinkse, and Porter (2003) where the authors normalize lots by tract size. We assume that $\tilde{c}_i^{(n)}$ are i.i.d..

By substituting (9) into the bidder's first order condition (4), dividing by $\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}$, explicitly writing out the various adaptations costs in K , and rearranging terms we can write

$$(10) \quad \frac{\theta_i^{(n)}}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}} = \frac{1}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}} \left(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{a,(n)} - \sum_{t=1}^T \frac{db_t^i(s^i)}{ds^i} q_t^{a,(n)} \left(\sum_{j \neq i} \frac{h_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})}{1 - H_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})} \right)^{-1} \right) + \frac{1}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}} \left[(1 - \tau_{a+}) A_+^{(n)} + (1 + \tau_{a-}) A_-^{(n)} + (1 - \tau_x) X^{(n)} + (1 + \tau_d) D^{(n)} \right] - \frac{1}{\bar{\mathbf{b}}^{(n)} \cdot \mathbf{q}^{a,(n)}} \left[P(\mathbf{b}^{i,(n)}) - \sum_{t=1}^T \frac{db_t^i(s^i)}{ds^i} \frac{\partial P(\mathbf{b}^{i,(n)})}{\partial b_t^i} \left(\sum_{j \neq i} \frac{h_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})}{1 - H_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})} \right)^{-1} \right].$$

³² Given the large number of bids in our data, we are able to estimate a separate empirical distribution for contracts where $\mathcal{N} = 2$ (using residuals from 266 bids), $\mathcal{N} = 3$ (552 bids), $\mathcal{N} = 4$ (671 bids), $\mathcal{N} = 5$ (639 bids), $\mathcal{N} = 6$ (444 bids), $\mathcal{N} = 7$ (448 bids), $\mathcal{N} = 8$ (200 bids), $\mathcal{N} = 9$ (180 bids), and $\mathcal{N} \geq 10$ (261 bids). This accounts for the total of 3,661 bids in our auctions.

³³ Additional details about the structural estimation procedure is provided in an online Appendix. Note that very similar results are obtained from using contract fixed effects or random effects in equation (8), as well as from including the number of bidders as an additional covariate in the random effects specification.

To complete our structural model we also include two additional sources of error in equation (10). The first source of error is that discussed in Section IVC, the potential endogeneity of ex post changes due to unobserved cost shocks. Recall that these omitted costs are observed by the firms, but not already accounted for in our cost estimate $\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}$. These costs may be correlated with $A_+^{(n)}$, $A_-^{(n)}$, $X^{(n)}$, and $D^{(n)}$ since projects with large ex post changes are likely to be more complicated and more expensive to complete. We will denote these additional unobserved costs as $\xi^{(n)}$.

The second source of error is an expectational error which results from bidders not having perfect foresight about $A_+^{(n)}$, $A_-^{(n)}$, $X^{(n)}$, and $D^{(n)}$. If risk neutral bidders have rational expectations then the FOC should simply be modified to include $EA_+^{(n)}$, $EA_-^{(n)}$, $EX^{(n)}$, and $ED^{(n)}$, the expected value of changes, instead of the actual values. In our data, we do not directly observe bidders' expectations. We therefore use well known strategies akin to those applied in Euler Equation estimation in macroeconomics and finance (described below) to estimate the model.³⁴ We denote the expectational error as $\omega^{(n)} \equiv (1 - \tau_{a+})(A_+^{(n)} - EA_+^{(n)}) + (1 + \tau_{a-})(A_-^{(n)} - EA_-^{(n)}) + (1 - \tau_x)(X^{(n)} - EX^{(n)}) + (1 + \tau_d)(D^{(n)} - ED^{(n)})$.

Given these two additional sources of error, $\xi^{(n)}$ and $\omega^{(n)}$, we can rewrite (10) as

$$(11) \quad \frac{\theta_i^{(n)}}{\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}} - \frac{\xi^{(n)}}{\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}} + \frac{\omega^{(n)}}{\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}} =$$

$$\frac{1}{\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}} \left(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{a,(n)} - \sum_{t=1}^T \frac{d b_t^i(s^i)}{d s^i} q_t^{a,(n)} \left(\sum_{j \neq i} \frac{h_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})}{1 - H_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})} \right)^{-1} \right)$$

$$+ \frac{1}{\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}} \left[(1 - \tau_{a+})A_+^{(n)} + (1 + \tau_{a-})A_-^{(n)} + (1 - \tau_x)X^{(n)} + (1 + \tau_d)D^{(n)} \right]$$

$$- \frac{1}{\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}} \left[P(\mathbf{b}^{i,(n)}) - \sum_{t=1}^T \frac{d b_t^i(s^i)}{d s^i} \frac{\partial P(\mathbf{b}^{i,(n)})}{\partial b_t^i} \left(\sum_{j \neq i} \frac{h_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})}{1 - H_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})} \right)^{-1} \right].$$

Equation (11) is identical to (10) except that we have brought over the two additional sources of error to the left-hand side. Taken together, we can define this composite error $\tilde{e}_i^{(n)}$:

$$\tilde{e}_i^{(n)}(\sigma, \tau_{a+}, \tau_{a-}, \tau_d, \tau_x, h, H) \equiv \frac{\theta_i^{(n)}}{\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}} - \frac{\xi^{(n)}}{\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}} + \frac{\omega^{(n)}}{\mathbf{b}^{(n)} \cdot \mathbf{q}^{a,(n)}}.$$

³⁴This strategy is used in the seminal paper by Hansen and Singleton (1982). We only observe the realized values of our variables and not a bidder's expectation of those values. This creates a bias similar to an errors in variables problem in linear regression. The idea in Hansen and Singleton is that information available to the agent at the time of bidding can be used to construct a valid instrument since the difference between the realized value of a random variable and its expected value must be orthogonal to current information. If this were not the case, an agent could improve her estimate of the expected value by conditioning on this instrument and hence would not be rational.

Letting $z_i^{(n)}$ denote the value of the instrument for bidder i in auction n , we will use (11) to form the moment condition below:

$$m_N(\sigma, \tau_{a+}, \tau_{a-}, \tau_d, \tau_x, h, H) = \frac{1}{N} \sum_n \sum_i \tilde{e}_i^{(n)}(\sigma, \tau_{a+}, \tau_{a-}, \tau_d, \tau_x, h, H) (z_i^{(n)} - \bar{z}_i^{(n)}).$$

As in our reduced form analysis, we use engineer identities as our primary instruments as they are uncorrelated with the unobserved cost shocks, $\xi^{(n)}$. They are also uncorrelated with the expectational errors, by definition, because the engineer is typically known at the time bidding occurs. We also include the engineer's cost estimate as an instrument because it is a natural shifter of the bidding strategies and thus, is correlated with the right-hand-side variables in (11). We index the moment condition by N to emphasize that the asymptotics of our problem depend on the number of auctions in our sample growing large.

Let $\hat{h}$ and $\hat{H}$ denote a first stage estimate of the bid densities and distributions. Let W be a positive semi-definite weight matrix. We use the following GMM estimator:

$$(\hat{\sigma}, \widehat{\tau_{a+}}, \widehat{\tau_{a-}}, \widehat{\tau_d}, \widehat{\tau_x}) = \arg \min m_N(\sigma, \tau_{a+}, \tau_{a-}, \tau_d, \tau_x, \hat{h}, \hat{H})' W m_N(\sigma, \tau_{a+}, \tau_{a-}, \tau_d, \tau_x, \hat{h}, \hat{H}).$$

The optimal weighting matrix can be calculated by using the inverse of the sample variance of $m_N(\cdot)$ from a first step estimate which uses the identity weighting matrix. Newey (1994) demonstrates that under suitable regularity conditions this estimator has normal asymptotics despite depending on a nonparametric first stage. Furthermore, the asymptotic variance surprisingly does not depend on how the nonparametric first stage is conducted, as long as it is consistent. The first stage estimates of $\hat{h}$ and $\hat{H}$ are quite precise given our regression coefficients.³⁵ Therefore, it is quite unlikely that our first stage bid density and distribution estimates introduce significant bias into the estimates.

Lastly, we need to specify a particular functional form for the skewing penalty function $P(\mathbf{b}^i)$. We use a convenient special case of the conditions we imposed on $P(\mathbf{b}^i)$ as follows:³⁶

$$(12) \quad P(\mathbf{b}^i | \bar{\mathbf{b}}) = \sigma \sum_{t=1}^T (b_t^i - \bar{b}_t)^2 \left| \frac{q_t^e - q_t^a}{q_t^e} \right|.$$

We chose this specification because the penalty increases for bids that are further away from the engineering estimate, and these get more weight when the actual quantity is further away from the estimated one. While we could consider a more flexible penalty function, the number of observations will limit the number of parameters we can include in this term. This, together with our objective of keeping

³⁵ Recall from footnote 31 that even when indexing these distributions by the number of bidders, each estimate is formed using between 180 and 671 individual bids.

³⁶ Strictly speaking, this does not guarantee that $\left. \frac{\partial P(\mathbf{b}^i | \bar{\mathbf{b}})}{\partial b_t^i} \right|_{b_t^i=0}$ is very large, but we still assume that an interior solution exists. The estimates act as a reasonable reality check.

the structure of the model as close to the standard literature as possible, is why we introduce this fairly parsimonious specification.

C. Estimating Markups

Given the estimates $(\hat{\sigma}, \widehat{\tau_{a+}}, \widehat{\tau_{a-}}, \widehat{\tau_d}, \widehat{\tau_x})$, we can recover an estimate of the contractors' implied markups. Using the functional form in (12) we estimate $\hat{\theta}^i$, contractor i 's total cost for installing the actual quantities by evaluating the empirical analogue of (4):

$$\begin{aligned} (\mathbf{b}^{i,(n)} - \hat{\mathbf{c}}^{i,(n)}) \cdot \mathbf{q}^{a,(n)} = & \frac{q_t^{a,(n)} - 2\hat{\sigma}(b_t^{i,(n)} - \bar{b}_t) \left| \frac{q_t^{a,(n)} - q_t^{e,(n)}}{q_t^{e,(n)}} \right|}{q_t^{e,(n)}} \left(\sum_{j \neq i} \frac{\hat{h}_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})}{1 - \hat{H}_j^{(n)}(\mathbf{b}^{i,(n)} \cdot \mathbf{q}^{e,(n)})} \right)^{-1} \\ & + \hat{\sigma} \sum_t (b_t^{i,(n)} - \bar{b}_t)^2 \left| \frac{q_t^{a,(n)} - q_t^{e,(n)}}{q_t^{e,(n)}} \right| - (1 - \widehat{\tau_{a+}}) A_+^{(n)} \\ & - (1 + \widehat{\tau_{a-}}) A_-^{(n)} - (1 - \widehat{\tau_x}) X^{(n)} - (1 + \widehat{\tau_d}) D^{(n)}. \end{aligned}$$

Using our estimates of $\hat{H}$, $\hat{h}$, $\hat{\sigma}$, $\widehat{\tau_{a+}}$, $\widehat{\tau_{a-}}$, $\widehat{\tau_x}$, and $\widehat{\tau_d}$, it is possible to evaluate the right-hand side of this equation.

D. Results

We summarize the structural estimates in Tables 8–11. Table 8 reports the parameter values from our semiparametric GMM estimator. The adaptation cost estimates are similar to the reduced form estimates discussed in Section IV. For instance, the last column of Table 8 implies that every dollar of positive adjustment generates an additional \$2.08 of adaptation costs, while a negative adjustment generates an additional \$2.42, though this later estimate is not statistically significant. Recall that our results control for the quantities that were actually installed by the contractor, $\mathbf{b} \cdot \mathbf{q}^a$. Moreover, as we described in the previous section, we have instrumented for the endogeneity of adjustments to account for a possible bias from remaining omitted cost variables. Therefore, we argue that this estimate reflects adaptation costs instead of omitted production costs $\xi^{(n)}$. It is worth noting that our reduced form estimates from Table 7 column 8 were smaller, at \$1.92 and \$0.55 for positive and negative adjustments respectively.

The other parameter estimates are also similar to the reduced form estimates. A dollar of extra work generates up to \$1.23 in adaptation costs, which almost matches the \$1.21 estimated in the reduced form specification. Deductions are estimated to generate \$1.49 in adaptation costs for every dollar of penalty assessed; however, the coefficient is not statistically significant at standard levels. As mentioned earlier in Section IV, the lack of significance is most likely a result of the small variation in this type of ex post change.

The estimated value of the skewing parameter, σ , is $-1.14E - 05$. Although the negative sign is inconsistent with the predictions of our theoretical model, this estimate is not statistically significant at standard levels. It is also extremely small

TABLE 8—STRUCTURAL ESTIMATION

	I-A	I-B	II-A	II-B
<i>Implied marginal transaction costs</i>				
Positive adjustments (τ^{A+})	4.254 (4.06)	1.968 (0.39)	4.35 (2.14)	2.081 (0.38)
Negative adjustments (τ^{A-})	0.815 (53.68)	0.944 (2.99)	2.574 (33.58)	2.418 (2.64)
Extra work (τ^X)*	1.078 (2.71)	1.079 (0.15)	1.928 (1.47)	1.226 (0.19)
Deductions (τ^D)	6.383 (77.74)	0.352 (4.78)	4.568 (39.96)	1.489 (3.58)
<i>Skewing parameter</i>				
Penalty (σ)	-4.19E-05 (5.91E-05)	-1.02E-05 (8.99E-06)	-4.66E-05 (3.52E-05)	-1.14E-05 (9.70E-06)
Observations	3,661	3,661	3,661	3,661
Weighting matrix**	Identity	Optimal	Identity	Optimal
Instruments used in second stage GMM		Resident engineer, engineer's estimate		Resident engineer, engineer's estimate, month and district dummies

Notes: * These estimates represent an upper bound on transaction costs associated with changes in scope. They do not account for marginal costs associated with performing the extra work, which for a reasonable profit margin of 20 percent would lower our estimate by \$0.80.

** Consistent GMM estimates were computed using the identity matrix as the weighting matrix. In a second step, efficient GMM estimates were computed using the optimal weighting matrix derived from the variance of the sample moments in the first step. Standard errors appear in parentheses.

in monetary terms and has no appreciable impact on profits or overall costs. The result that there are no large penalties from skewing is quite robust to alternative specifications for the functional form of the skewing penalty. However, recall from Section I that positive and negative adjustments are essentially due to renegotiating unit prices. As an empirical matter, it may be difficult to separately identify a quadratic effect of overruns and underruns, as captured in σ , from the linear effect captured in τ_{a+} and τ_{a-} .

In Tables 9 and 10, we summarize our estimates of bidders' markups. Our results suggest that the industry is quite competitive. The median profit margin is 3.8 percent for all bids and 12 percent for winning bids. We note that Granite Construction Inc., the largest bidder in our data, is a publicly traded company and reported a net profit margin of 3.1 percent in 2005. The construction industry average according to Standard and Poors is 1.9 percent. Profit margins based on SEC filings and our conception of profits differ in many respects. However, the available direct evidence on profit margins suggests that the construction industry is quite competitive and our results are consistent with this evidence.

As Table 9 demonstrates, markups over direct costs $(\mathbf{b}^i - \mathbf{c}^i) \cdot \mathbf{q}^a$ are considerably higher than the profit margin. This is because our analysis distinguished between the direct costs of completing the project *without adaptation costs* and the added adaptation costs. The median markup over direct costs, $(\mathbf{b}^i - \mathbf{c}^i) \cdot \mathbf{q}^a$ is \$88,051 for all bids and \$230,682 for winning bids. The ratio of the markup over direct costs to the cost estimate for the median job is 8.1 percent for all bids and 17.7 percent

TABLE 9—MARKUP DECOMPOSITION

	All bidders			Winning bidders only		
	Bottom tenth percentile	Median	Top tenth percentile	Bottom tenth percentile	Median	Top tenth percentile
Direct markup: $(b_i - c_i)q^a$	12,087	88,051	835,060	38,938	230,682	1,479,238
Direct markup/estimate	2.9%	8.1%	26.1%	7.5%	17.7%	59.8%
Ex post changes: $A + X + D - TC(A, X, D)$	-375,140	-29,524	-907	-396,952	-34,439	-972
Ex post changes/estimate	-11.6%	-2.9%	-0.2%	-12.2%	-3.1%	-0.2%
Skewing penalty: $\sigma \cdot \Sigma_i [(b_i - b)^2 \%Over_i]$	-143.52	-0.78	-0.00	-61.38	-0.27	0.00
Skewing penalty/estimate	-0.0095%	-0.0001%	0.0000%	-0.0031%	0.0000%	0.0000%
Total profit: Π	7,214	44,903	403,496	29,984	159,683	1,108,879
Total profit/estimate	1.5%	3.8%	15.7%	5.1%	12.0%	53.3%

TABLE 10—MARKUPS IMPLIED BY STANDARD MODEL WITHOUT ADAPTATION COSTS OR EX POST CHANGES

	All bidders			Winning bidders only		
	Bottom tenth percentile	Median	Top tenth percentile	Bottom tenth percentile	Median	Top tenth percentile
Direct markup: $(b_i - c_i)q^a$	7,141	44,443	405,350	29,628	156,691	1,096,923
Direct markup/estimate	1.6%	3.7%	15.7%	5.0%	11.8%	53.5%

for winning bids. Contractors are effectively extracting high margins on the components of the contract specified at the time of bidding, in order to cover the costs of adapting to changes in the contract ex post.

In Table 10, we compare the estimates in Table 9 with the estimated markups found using more standard methods that ignore the ex post changes to quantities and payments. Using our first stage estimates of $\hat{H}_j(\mathbf{b}^i \cdot \mathbf{q}^e)$ and $\hat{h}_j(\mathbf{b}^i \cdot \mathbf{q}^e)$, we evaluate the empirical analogue of equation (5), which is essentially the estimator discussed in Guerre, Perrigne, and Vuong (2000). Equation (5) is a special case of our model when we assume that $\mathbf{q}^a = \mathbf{q}^e$ and that ex post changes can be excluded from the FOCs. Previous empirical structural models of first price procurement auctions make these assumptions. Using this approach, the median total markup is \$156,691, or 11.8 percent of the estimate for winning bidders. These results almost exactly match the median profit margins reported in the last two rows of Table 9. By comparing the FOCs of the two models, this should not be surprising. The only difference in the *net profit margin* under the two approaches comes from the skewing penalty and the discrepancies between estimated and actual item quantities. Ex post changes will shift the bid, changing what we refer to as the direct markup, but do not alter the contractors' net profit margins. This is an important observation to the extent that it is a consequence of the "pass through" of costs: the profit margins over total costs are practically the same in both empirical models. However, our approach distinguishes the direct costs from the adaptation costs that follow from incompletely specified contracts. Our results therefore suggest that the standard

TABLE 11—ADAPTATION COSTS

Percentile	Total adaptation costs	As a fraction of contract's estimate	As a fraction of contract's estimated ex post profit
<i>Panel A. Upper bound</i>			
Bottom tenth	\$4,581 [−3,807, 6,663]	0.90% [−0.3%, 1.4%]	3.30% [−1.1%, 4.9%]
Median	\$100,978 [41,255, 147,183]	8.4% [4.3%, 13.1%]	58.5% [29.2%, 91.0%]
Top tenth	\$1,074,675 [680,775, 1,518,007]	31.5% [19.2%, 45.7%]	323.4% [213.9%, 473.6%]
<i>Panel B. Lower bound</i>			
Bottom tenth	\$972 [−61,362, 2,621]	0.2% [−5.0%, 0.5%]	0.7% [−39.4%, 2.0%]
Median	\$38,385 [−532, 90,467]	3.80% [−0.1%, 8.1%]	25.10% [−0.4%, 56.9%]
Top tenth	\$656,088 [267,711, 1,049,892]	16.6% [7.0%, 32.0%]	173.0% [70.7%, 327.6%]

Notes: The total adaptation cost is calculated as $2.081(A_+) + 2.418|A_-| + 1.226(X) + 1.489|D|$. We consider this to be an upper bound because it attributes the coefficient on extra work to pure transaction costs, rather than marginal costs of production. This amounts to an assumption of a 100 percent profit margin on extra work, making it analogous to positive adjustments in compensation. A more conservative approach is presented in the bottom panel. In the extreme, firms may perform extra work at a zero profit margin, meaning each dollar awarded through a change in scope just covers the cost of performing that work. With \$1 of marginal costs for every \$1 of extra work, that still leaves approximately \$0.23 to be explained by transaction or adaptation costs. Therefore we calculate a lower bound on adaptation costs as $2.081(A_+) + 2.418|A_-| + 0.226(X) + 1.489|D|$.

The 95 percent confidence intervals, appearing in brackets, were calculated by taking the lower and upper confidence bounds of the parameter estimates and using them in the transaction cost equation instead of the point estimates.

FOC used in previous empirical studies is misspecified because it does not account for ex post changes. In our application, failing to account for contract adaptations leads to estimates with a very different economic interpretation, and as discussed below, with very different policy implications.

Table 11 reports lower and upper bounds for the total adaptation costs estimated on each project. These bounds are determined based on the possible margins that firms may collect on extra work through change orders. Unlike adjustments and deductions, payments made on extra work are not purely monetary transfers but are negotiated to cover costs of additional work that cannot be fully accounted for in the itemized quantities listed in the contract.³⁷ Suppose that the contractor was able to earn a profit margin from renegotiating these changes. As an upper bound, we can consider a case where contractors earn an extra \$1 in profits for every \$1 of changes in scope, X . Then the entire $1.23X$ would be attributable to adaptation costs. Our upper bound for the total estimate of adaptations costs would therefore be calculated as $2.091A_+ + 2.418|A_-| + 1.226X + 1.489|D|$. A much more conservative approach would be to assume firms make zero margin on extra work, i.e., that every \$1 received ex post merely goes toward covering the costs of performance.³⁸ Even

³⁷ Industry sources suggest that a 20 percent profit margin on change orders is most common. It is helpful to recall that our cost estimate controls for the quantities actually installed and that positive and negative adjustments are effectively changes to compensation from the unit prices. Hence, these are a pure transfer and do not involve additional costs that we have not controlled for in our cost estimate.

³⁸ This makes the reasonable assumption that firms do not negotiate ex post changes that result in them losing money.

in this case, there remains an additional \$0.23 that cannot be attributed to production costs, and we instead attribute to the costs of adaptation. The lower bound is therefore calculated as $2.091A_+ + 2.418|A_-| + 0.226X + 1.489|D|$. The median estimate of adaptation costs is a sizable component of costs by any standard. It has a lower bound of 3.8 (−0.1, 1) percent of the engineer's estimate and an upper bound of 8.4 (3, 13.1) percent of that estimate.

VI. Implications for Government Procurement

Our analysis offers some lessons for the design of procurement auctions. The first is that the existing system does a good job of limiting rents and promoting competition. The *total* markup is modest, with the median bidder in our sample expecting a profit of 3.8 percent of the estimate. More interesting, though, is how firms make their markup. Because adaptation costs erode more than any positive gains from change orders, firms increase their bids *ex ante* to extract high rents on prespecified project items. Among winning bidders the median value of this direct markup, $(b_i - c_i) \cdot q^a$, is 17.7 percent of the project's estimated costs.

Second, both the reduced form and structural estimates imply that adaptation costs are substantial. The implied adaptation costs range from 55 cents to around two dollars for every dollar in change. These numbers might be surprising to economists who emphasize private information and moral hazard as the main sources of inefficiency. However, the results are consistent with current thinking in Construction and Engineering Project Management. (See, Bartholomew 1998; Clough and Sears 1994; Hinze 1993; and Sweet 1994. Also see Bajari and Tadelis 2001 for a more complete set of references and discussion of the literature.) One of the central concerns emphasized in this literature are methods for minimizing the costs of disputes between contractors and buyers. The topic of controlling contractor margins, by comparison, receives relatively little emphasis in this literature.

The bounds on adaptation costs that were derived in Section VD suggest that Caltrans spent on average \$362,293 to \$538,549 per contract on adaptations costs during, or a *total* of \$297 million to \$441 million on adaptation costs during our five and a half year sample period. The average ratio of adaptation costs to the winning bid is 0.075 to 0.141. Even half of our lower bound would be substantial. An implication of competition is that Caltrans, and hence the taxpayer, is ultimately responsible for expected adaptation costs on the project as they are directly passed on from the bidders. Of course, we have no way of estimating the direct adaptation costs born by Caltrans in the form of legal fees and other haggling costs, which probably add significant costs. Since the source of these costs is the incompleteness of project design and specifications, an obvious policy implication is to consider increasing the *ex ante* costs and efforts put into estimating and specifying projects before they are let out for bidding. Our analysis does not allow us to speculate on the costs and benefits of adding more engineering efforts *ex ante*, and it would be impossible to avoid contractual incompleteness and the implied agency problems altogether. However, since the magnitude of adaptation costs is sizable, there is room to consider some experimentation with more careful and costly design efforts, and to carefully examine the results of any such added effort in *ex ante* engineering.

In related work, Bajari, McMillan, and Tadelis (2009) study how private sector, nonresidential construction contracts are awarded in Northern California between 1995 and 2000. Unlike the public sector, private sector buyers can easily use mechanisms other than competitive bidding to select a contractor. They find that open competitive bidding is only used in 18 percent of the contracts and that 44 percent of the contracts are negotiated. Also, negotiated contracts are more commonly used for projects that appear to be the most complex for which ex post changes in plans and specifications are likely.

A perceived advantage of negotiated contracts is that they allow the architect, buyer, and contractor to discuss the project plans so that the contractor can point out pitfalls and suggest modifications to the project design before work begins. Furthermore, negotiated contracts are often based on cost plus compensation. As discussed in Bajari and Tadelis (2001), the poor incentives to control costs that cost plus contracts provide are compensated for by the ease in which renegotiating terms is achieved. When changes occur, the contractor presents his receipts for the additional expenses and is reimbursed, thus avoiding the often acrimonious process of negotiating change orders and reducing adaptation costs. The results of this paper complement these previous papers by confirming that adaptation costs are high in the construction industry, so that economizing on ex post adaptation costs is an important potential source of cost savings. This may outweigh the benefits of competitive bidding in selecting the lowest cost contractor, and providing the contractor with strong incentives to lower production costs.

In the public sector, the use of negotiated contracts is problematic. Allowing for greater discretion in contractor selection increases the possibility for favoritism, kick backs, and political corruption. The competitive bidding system is less prone to corruption since it allows for free entry by qualified bidders and there is an objective criteria for selecting the winning bidder. An important policy issue is whether it is possible to construct a mechanism that balances ex post adaptation costs with the potential for corruption. To the best of our knowledge, this question has not been explored in the existing theoretical literature. Our research suggests that developing such a mechanism could dramatically improve efficiency in public sector procurement.

REFERENCES

- Asker, John, and Estelle Cantillon. 2008. "Properties of Scoring Auctions." *RAND Journal of Economics* 39 (1): 69–85.
- Athey, Susan, and Philip A. Haile. 2006. "Empirical Models of Auctions." In *Advances in Economics and Econometrics, Theory and Applications: Ninth World Congress of the Econometric Society*. Volume 2, edited by Richard Blundell, Whitney Newey, and Torsten Persson, 1–45. New York: Cambridge University Press.
- Athey, Susan, and Philip Haile. 2007. "Nonparametric Approaches to Auctions." In *Handbook of Econometrics*. Vol. 6A, edited by James Heckman and Edward Leamer, 3847–965. Amsterdam: Elsevier Science, North-Holland.
- Athey, Susan, and Jonathan Levin. 2001. "Information and Competition in U.S. Forest Service Timber Auctions." *Journal of Political Economy* 109 (2): 375–417.
- Athey, Susan, Jonathan Levin, and Enrique Seira. 2011. "Comparing Open and Sealed Bid Auctions: Evidence from Timber Auctions." *Quarterly Journal of Economics* 126 (1): 207–57.
- Bajari, Patrick, Stephanie Houghton, and Steven Tadelis. 2014. "Bidding for Incomplete Contracts: An Empirical Analysis of Adaptation Costs: Dataset." *American Economic Review*. <http://dx.doi.org/10.1257/aer.104.4.1288>.

- Bajari, Patrick, Robert McMillan, and Steven Tadelis.** 2009. "Auctions versus Negotiations in Procurement: An Empirical Analysis." *Journal of Law, Economics, and Organization* 25 (2): 372–99.
- Bajari, Patrick, and Steven Tadelis.** 2001. "Incentives versus Transaction Costs: A Theory of Procurement Contracts." *RAND Journal of Economics* 32 (3): 387–407.
- Bajari, Patrick, and Lixin Ye.** 2003. "Deciding between Competition and Collusion." *Review of Economics and Statistics* 85 (4): 971–89.
- Bartholomew, Stuart H.** 1998. *Construction Contracting: Business and Legal Principles*. Upper Saddle River, NJ: Prentice-Hall, Inc.
- California Department of Transportation.** 1999–2005. "Major Construction Payment & Information System. Caltrans Contract Listing by District." Division of Accounting, hardcopies and electronic files <http://www.dot.ca.gov/hq/asc/oap/payments/public/ctnums.htm> (accessed August 2004–July 2009).
- California Department of Transportation.** 1999–2005. "Contract Cost Data Book." Division of Engineering Services–Office Engineer <http://dot.ca.gov/hq/esc/oe/awards/index.html> (accessed August 2004–July 2009).
- California Department of Transportation.** 1999–2005. "Caltrans Bid Summary File Library Site." Division of Engineering Services–Office Engineer, hardcopies and electronic files http://www.dot.ca.gov/hq/esc/oe/awards/bidsum_html/ (accessed August 2004–July 2009).
- Chakravarty, Surajeet, and W. Bentley MacLeod.** 2009. "Contracting in the Shadow of the Law." *RAND Journal of Economics* 40 (3): 533–57.
- Che, Yeon-Koo.** 1993. "Design Competition through Multidimensional Auctions." *RAND Journal of Economics* 24 (4): 668–80.
- Clough, Richard H., and Glenn A. Sears.** 1994. *Construction Contracting*. 6th ed. New York: Wiley.
- Crocker, Keith J., and Kenneth J. Reynolds.** 1993. "The Efficiency of Incomplete Contracts: An Empirical Analysis of Air Force Engine Procurement." *RAND Journal of Economics* 24 (1): 126–46.
- Elyakime, Bernard, Jean-Jacques Laffont, Patrice Loisel, and Quang Vuong.** 1994. "First-Price Sealed-Bid Auctions with Secret Reservation Prices." *Annales d'Economie et de Statistique* 34 (1): 115–41.
- Guerre, Emmanuel, Isabelle Perrigne, and Quang Vuong.** 2000. "Optimal Nonparametric Estimation of First-Price Auctions." *Econometrica* 68 (3): 525–74.
- Haile, Philip A.** 2001. "Auctions with Resale Markets: An Application to U.S. Forest Service Timber Sales." *American Economic Review* 91 (3): 399–427.
- Haile, Philip A., Han Hong, and Matthew Shum.** 2003. "Nonparametric Tests for Common Values in First-Price Sealed-Bid Auctions." National Bureau of Economic Research Working Paper 10105.
- Hanna, Awad S., and Murat Gunduz.** 2004. "Impact of Change Orders on Small Labor-Intensive Projects." *Journal of Construction Engineering and Management* 130 (5): 726–33.
- Hansen, Lars Peter, and Kenneth J. Singleton.** 1982. "Generalized Instrumental Variables Estimation of Nonlinear Rational Expectations Models." *Econometrica* 50 (5): 1269–86.
- Hendricks, Kenneth, Joris Pinkse, and Robert H. Porter.** 2003. "Empirical Implications of Equilibrium Bidding in First-Price, Symmetric, Common Value Auctions." *Review of Economic Studies* 70 (1): 115–45.
- Hinze, Jimmie W.** 1993. *Construction Contracts*. McGraw-Hill Series in Construction Engineering and Project Management. Boston: Irwin/McGraw-Hill.
- Jofre-Bonet, Mireia, and Martin Pesendorfer.** 2003. "Estimation of a Dynamic Auction Game." *Econometrica* 71 (5): 1443–89.
- Krasnokutskaya, Elena.** 2011. "Identification and Estimation of Auction Models with Unobserved Heterogeneity." *Review of Economic Studies* 78 (1): 293–327.
- Lebrun, Bernard.** 2006. "Uniqueness of the Equilibrium in First-Price Auctions." *Games and Economic Behavior* 55 (1): 131–51.
- Newey, Whitney K.** 1994. "The Asymptotic Variance of Semiparametric Estimators." *Econometrica* 62 (6): 1349–82.
- Paarsch, Harry J.** 1992. "Deciding between the Common and Private Value Paradigms in Empirical Models of Auctions." *Journal of Econometrics* 51 (1–2): 191–215.
- Porter, Robert H., and J. Douglas Zona.** 1993. "Detection of Bid Rigging in Procurement Auctions." *Journal of Political Economy* 101 (3): 518–38.
- Silverman, Bernard W.** 1986. *Density Estimation for Statistics and Data Analysis*. London: Chapman and Hall.
- Sweet, Justin.** 1994. *Legal Aspects of Architecture, Engineering and the Construction Process*. 5th ed. Minneapolis-St. Paul, MN: West Publishing Company.
- Williamson, Oliver E.** 1971. "The Vertical Integration of Production: Market Failure Considerations." *American Economic Review* 61 (2): 112–23.

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